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Computations of probable maximum precipitation estimates

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1. Introduction

Probable maximum precipitation (PMP) is defined by the World Meteorological Organization as “the greatest depth of precipitation for a given duration meteorologically possible for a design watershed or a given storm area at a particular location and time of year, with no allowance made for long-term climatic trends.” An overview of the conventional PMP estimation methods is provided by WMO (2009a). In the course of scientific applications, PMP estimation methods have continuously evolved and improved. This is particularly the case when it comes to estimate the spatiotemporal PMP distribution (Beauchamp et al., 2013; Felder and Weingartner, 2016; Lagos-Zúñiga and Vargas, 2014), accounting for climate change and stationarity issues (Rousseau et al., 2014; Stratz and Hossain, 2014), and incorporating uncertainty bands (Micovic et al., 2015; Faulkner et al., 2016; Rouhani and Leconte, 2016).

In the past century, numerous water infrastructures have been built to facilitate irrigation, hydropower generation, transportation, and municipal water use. In a changing climate, extreme precipitation events are projected to be more frequent and intense, exceeding known historical records (Kunkel et al., 2013). Along with structural safety, the hydrologic safety of water infrastructures is now therefore gaining more attention, since overtopping or embankment failure would bring catastrophic human and societal loss (Lane, 2013). Most of the water infrastructures, especially the hazardous ones located upstream of population centers, are often designed considering the standard PMP (Hossain et al., 2012).

Information of the PMP is necessary for designing high-hazard structures such as large dams and spillways, flood protection works, levees, and nuclear power plants (Singh et al., 2018). A common practice for determining the design flood for large scale dams is to estimate PMP and then transform it into the probable maximum flood (PMF). PMF is used to size the hydraulic structure such that risk of their failure is minimized, and it may be generated from using rainfall-runoff relationship, hydraulic, and hydrological model. PMF is the theoretical maximum flood that is expected to occur at a particular location due to the most severe combination of critical meteorological and hydrological conditions.

Dams classified in the “extreme hazard” category are those that, in case of failure, have the potential to cause human loss of life and severe environmental and financial losses. Consequently, they have to meet strict design requirements regarding the maximum flow the dam must be able to pass based on the concept of PMF. As mentioned in WMO (2009a), PMP calculations depend on data and should always be presented as approximations, since the value depends fundamentally on the amount and quality of the data available and the depth of analysis.

The possible change in PMP will have some serious implications on such structures due to their very long-life span (>100–500 years) and may alter the risk and reliability associated with such structures, designed against a stationary estimate of PMP (Kunkel et al., 2013). Hence, the study on temporal change of PMP is one of utmost importance in the context of climate change (Sarkar and Maity, 2020).

1.1 Background and importance of PMP estimations

The frequency and intensities of extreme rainfalls and record-breaking floods are observed to be increasing due to extreme rainfall events (Maleki and Eslamian, 2022). Floods are becoming more apparent due to design limits of flood defense structures (storm drainage, embankments, dams, etc.) exceeded. The risk of incidence of flood was due to variations occurrence as a

result of global warming which brings a rise in temperature causing heavy rain and rise in sea level (Zaidi et al., 2014). In 2014, the East Coast of Peninsular Malaysia has experienced the worst flood events since 1971 that caused massive destruction of land use (Rahaman, 2014). During that time, Kelantan flood known as the big yellow flood experiences ARIs near and over 100 years recorded at many stations upstream of the Kelantan river basin. Enhanced rainfall was experienced due to the combined effect of the monsoon season, Madden Julian Oscillation, and temperature below anomalies at the Siberian High (Alias et al., 2016).

Another event in 2016 Louisiana flood caused by a severe weather system which brought prolonged rainfall in about 72h, resulting in widespread and catastrophic flooding (Wang et al., 2019). The state of Kerala (India) witnessed large-scale flooding in 2018 due to unprecedented extreme rainfall (above normal seasonal rainfall) and almost full reservoirs resulted in a significant release of water in a short span of time, which affected millions of people and caused 400 or more deaths (Mishra and Shah, 2018).

Globally, increases in heavy precipitation have been reported in many regions, including U.S. through the first part of the 21st century (Field et al., 2012; Seneviratne et al., 2012). Further, extreme precipitation events show statistically significant trends in North America (Guilbert et al., 2015; Roque-Malo and Kumar, 2017). It is generally agreed that climate change does have an impact on severe precipitation extremes, and its effect should therefore be considered (Simonović, 2012). The situation under extreme events may worsen in the future (Field et al., 2012). Papalexiou and Montanari (2019) found that there was an increasing trend in the frequency of daily precipitation extremes that were highly unlikely under the assumption of stationarity when the global warming accelerated.

This shows the importance of a reliable and up-to-date return period estimation for the designing of hydraulics and flood protection structures especially in this current changing climate. The existence of extreme rainfall values or outliers can affect the quality of the quantiles estimated when using a conventional frequency analysis during the design stages of hydraulics and flood defense structures. This is due to the extreme values which were not fitted to the frequency analysis model. In future, this problem would be more significant for many regions which receive higher rainfall intensities than the normal amount. The consideration of using PMP is important to be ready for extreme events. Many strongly believe that the PMP can indeed be used to represent maximum rainfall values or representing the upper boundary of the extreme rainfall frequency models (Desa et al., 2001; Rezacova et al., 2005). Even so, PMP estimates are location dependent and sensitive to the available dataset (Lee and Singh, 2020). As the available data and our understanding increase, it may be appropriate to refine and adjust PMP estimates with regional or site-specific studies (Zhang et al., 2019).

The fifth IPCC report also showed that future air temperature and humidity are expected to gradually increase (IPCC, 2015; Jakob and Walland, 2016). Thus, climate change will have a strong impact on moisture maximization and PMP as a result (Thuya et al., 2019a). Recent studies have suggested that PMP is likely to rise in most areas of the world under a warmer climate (Beauchamp et al., 2013; Lee, 2017; Rastogi et al., 2017; Rousseau et al., 2014). Though most developed countries have well maintained long-term hydro-climatic data, difficulties in data availability are always a major issue hindering many studies in developing nations such as Vietnam (Thuya et al., 2019a). In addition, the magnitudes of the maximum precipitable water could rise with increasing air temperature and thus could magnify extreme rainfall in the future (Ishida et al., 2018; Kunkel et al., 2013).

Given the considerable uncertainties that may influence PMP estimates, providing a range of PMP values by evaluating the uncertainty of values, that is the range of values that might be possible under equivalent conditions, rather than relying only on single point estimates is necessary and more suitable, but is often ignored in the literature. According to Alaya et al. (2018), only a few studies have dealt with this subject. For instance, Salas et al. (2014) provides an uncertainty estimate using a statistical method, where the expected value and variance of the calculated PMP value are obtained by assuming that the annual maximum of 24-h precipitation is Gumbel distributed.

2. Methodology of PMP estimation

The World Meteorological Organization (WMO) recognizes several methodologies to estimate PMP (WMO, 2009a). The two main methods are: (1) a statistical approach known as Hershfield statistical method and (2) the physical approach known as the hydrometeorological method. Depending on the availability and limitations of data, both methods are accepted and are widely used. According to WMO (2009a), more reliable PMP estimates can be calculated via the hydro-meteorological method (physical method) if various meteorological data are available. This is a significant problem for those who have difficulties to acquire the data. A reliable PMP estimates using the statistical method require long observation data.

2.1 Physical method

The storm maximization approach (also referred to as the storm separation method in Hydrometeorological Report, HMRs (Hansen, 1994; Corrigan, 1999), which consists of moisture maximization and storm transposition, is the most adopted and

extensively studied approach (Casas et al., 2011; Rouhani and Leconte, 2016; Hansen et al., 1994). Through this orographic and convergence rainfall amounts can be computed. The convergence rainfall is referred to as the free-atmospheric forced precipitation, FAFP (Hansen, 1994). Storm events which occurred for a specified duration d (1, 2, 3, 4, or 5 days) are identified from the historical record of precipitation. Let N_d denote the number of events having duration d . The estimate of maximized precipitation $P_{max}^{(e)}$ is determined corresponding to each of the events e ($= 1, \dots, N_d$), and the largest of those estimates is declared as PMP corresponding to duration d .

$$\text{PMP} = \frac{\text{Max}}{\forall e} \left[P_{max}^{(e)} \right] = \frac{\text{Max}}{\forall e} \left[P^{(e)} \times \frac{W_{max}^{(e)}}{W^{(e)}} \right] \quad (1)$$

where $P^{(e)}$ is the precipitation observed during a storm event e , $W^{(e)}$ denotes precipitable water in the air column above the target location during the storm event, and $W_{max}^{(e)}$ represents the maximized precipitable water in the air column for the month in which the storm event has occurred. If events have duration greater than 1 day, the month and season of occurrence of the event are decided based on the day recording the maximum depth of precipitation. The estimate of $W^{(e)}$ can be obtained using the following equation (Smith, 1966).

$$W^{(e)} = \exp [0.1133 - \ln (\lambda_m + 1) + 0.0393T_d] \quad (2)$$

where λ_m represents exponent in power law relationship assumed between moisture profile and height of the atmospheric column. Recently, Chen and Bradley (2006) have suggested that power law moisture profile suggested by Smith (1966) could be a reasonable choice for representing the relationship between the precipitable water and persisting surface dew point temperature T_d . A researcher provided estimates of λ_m for various latitudinal ranges over the Northern hemisphere for different seasons (winter, spring, summer, and fall) in which a storm could occur. The estimate of T_d (in degrees Celsius) corresponding to the day of storm event is computed as,

$$T_d = \frac{237.3 \ln (R_h) + 237.3 \left(\frac{17.27T_s}{237.3 + T_s} \right)}{17.27 - \ln (R_h) - \left(\frac{17.27T_s}{237.3 + T_s} \right)} \quad (3)$$

where R_h is the surface relative humidity, and T_s represents the maximum surface temperature (in degrees Celsius) on the day of the storm. When the duration of precipitation exceeds 1 day, T_d for the storm could be computed as the arithmetic average of T_d values corresponding to all the days over which the storm has occurred. The estimate of $W_{max}^{(e)}$ considered in Eq. (1) can be obtained using Eq. (2) by replacing T_d with the maximum dew point temperature T_d^{max} for the month in which the storm has occurred. When observations of T_d are available for at least 50 years, T_d^{max} is considered the highest recorded T_d value for the month in which the storm has occurred. Otherwise T_d^{max} is considered the quantile estimate of T_d corresponding to 100-year return period (or 0.99 nonexceedance probability), which can be determined by fitting a frequency distribution to historical monthly maximum values of T_d for the month in which the storm has occurred (WMO, 1986). Appropriate frequency distribution could be identified using Kolmogorov-Smirnov goodness-of fit test. Formulations which are useful for quantile estimation in the case of various frequency distributions can be found in Hamed and Rao (2019).

2.2 Statistical method

Most PMP estimation procedures are based on meteorological analyses. However, sometimes site-specific meteorological data are not available. Statistical methods are useful in such cases (Fattahi et al., 2011). The most widely used statistical estimation method developed by Hershfield (1961, 1965a, b), has been adopted by the WMO (2009b) as one of the standard methods for PMP estimation. This method takes into account actual historical data at a particular location and expresses it in terms of statistical parameters (Koutsoyiannis, 1999). The procedure as developed by Hershfield (1961, 1965a, b) is based on the general frequency equation by Chow (1951) in the form of

$$X_T = \bar{X}_n + K S_n \quad (4)$$

where X_T is the rainfall for return period T , $\bar{X}_n$, and S_n are respectively, the mean and standard deviation of a series of n annual maxima, and K is a common statistical variable, which varies with the different frequency distributions fitting extreme value hydrological data. If the maximum observed rainfall X_m is substituted for X_T , and K_m for K , K_m is then the number of standard deviations to be added to obtain X_m or (Fig. 1)

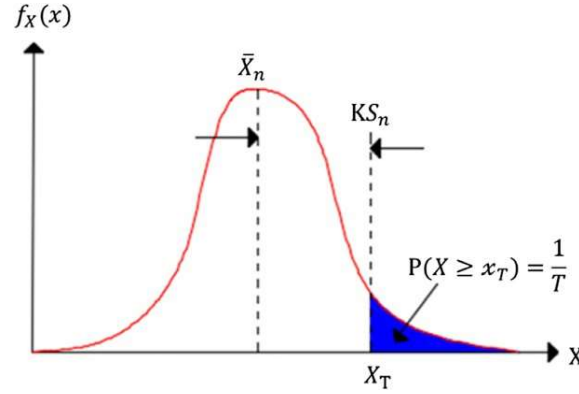


FIG. 1 Sketch describing the general frequency equation by Chow (1951).

$$X_m = \bar{X}_n + K_m S_n \quad (5)$$

The equations used for estimating the PMP value for a station are as follows:

$$X_{PMP} = \bar{X}_n + S_n \times K_m \quad (6)$$

$$K_m = \frac{X_{max} - \bar{X}_{n-1}}{S_{n-1}} \quad (7)$$

where X_{PMP} is the PMP estimates for a station, $\bar{X}_n$ is the mean of the annual extreme series, S_n is the standard deviation of the annual extreme series, K_m is the frequency factor which depends on the availability of data period, X_{max} is the highest rainfall value at the station, $\bar{X}_{n-1}$ is the mean of the annual extreme series without the largest value, and S_{n-1} is the standard deviation of the annual extreme series without the largest value.

First, the parameters $\bar{X}_n$, S_n , and K_m are calculated using Eq. (7). Then, K_m values for all stations are plotted against the $\bar{X}_n$ values respectively and an envelope curve is drawn. The new K_m value is picked up from the envelope line for each station's $\bar{X}_n$. The envelop of the K_m values are done in various ways. According to WMO (2009a), initially the enveloping value of K_m was based on records of 24-h rainfall of 2700 stations; however, it was later found that the K_m values vary inversely with the mean of the rainfall series. It is also suggested that the maximum value of 15 in their study may be too high for areas of generally heavy rainfall and too low for arid areas. An envelopment of the K_m values is presented in the WMO (2009a) manual. A case-study in Malaysia, where the rainfall is generally high (annual rainfall of around 2000–2500 mm), Desa (2001) found that the K_m value of 15 was too high. Using local Malaysia data and by using rainfall data of selected stations within site specific boundaries, Desa (2001) found that the highest K_m value in Malaysia is between 7.5 and 9.0 (Desa et al., 2001; Desa and Rakhecha, 2007). This is supported as well in another study by Sidek et al. (2013). Table 1 shows a variety of K_m values in other countries. A study in Vietnam shows K_m values within a range of 4–12, Korea has an overall value of less than 4 but with a highest of 6, India has values between 0 and 16, and Iran a variable climate of semiarid and subtropical climate has a value of between 2.5 and 5.3. The PMP values for each station are then computed using Eq. (6) by replacing K_m with the new transposed value or the highest K_m value. Fig. 2 shows a summary of the flow on estimating the Hershfield statistical PMP.

2.3 Multifractal approach

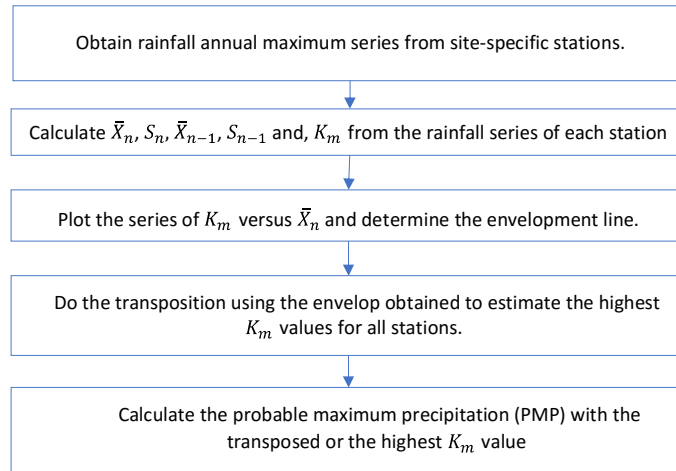
It is an approach that considers precipitation intensity as a multifractal field and describes properties of the field at various temporal scales by using a scale-invariant multiplicative cascade model (Lavallée, 1991; Olsson, 1995; Lovejoy and Schertzer, 2007). Multifractal scaling properties can be described by the following equations (Lovejoy et al., 1987):

$$\begin{aligned} \varepsilon_\lambda &\sim \lambda^\gamma \text{ for } \lambda > 1 \\ \Pr(\varepsilon_\lambda > \lambda^\gamma) &\sim \lambda^{-c(\gamma)} \end{aligned} \quad (8)$$

where ε_λ is the intensity of precipitation at scale ratio $\lambda (=L/\tau)$; L represents length of precipitation time series, τ is the time scale of interest (e.g., 1 day, 2 days); γ is referred to as the order of singularity, and its probability distribution is described by

TABLE 1 K_m values from Hershfield statistical method.

Location	Year	K_m values of Hershfield method
Selangor, Malaysia	Desa et al. (2001)	1-day rainfall duration, K_m values 2–9
Johor, Malaysia	Desa and Rakhecha (2007)	1-day rainfall duration, K_m values 2–7.5
Cameron Highland, Pahang, Malaysia	Sidek et al. (2013)	K_m values 2–8 for 1-day rainfall, 1–6.5 for 3-day rainfall and 1–5.9 for 5-day rainfall
Bac Kan, Thai Nguyen and Tuyen Quang, Vietnam	Thuya et al. (2019b)	24-h precipitation, K_m values are 4.58–5.63 with an outlier of 12.78
South Korea	Lee (2017)	1-month rainfall duration, K_m values are less than 4.0 with highest value of 6.03
Indian mainland, India	Subharthi Sarkar (2020a, b)	1-day rainfall duration (1901 to 2000), K_m values ranges from 0 to 16 and new method is 16.7
Mahanadi and Godavari River basin, India	Srinivas (2021)	2-day rainfall duration, K_m values ranges from 0 to 12
Qareh-Su basin, Iran	Zahra Afzali-Gorouh (2018)	1-day rainfall duration, K_m values were found to vary between 17 and 18. After using envelope method, new K_m value is ranged from 2.2 to 5.3
Amur River basin, Russia	Bolgov (2019)	1-day rainfall duration, K_m values are 0.8–2.8

**FIG. 2** Hershfield statistical procedure as recommended by WMO (2009a).

convex codimension function $c(\gamma)$, which measures the fraction of the probability space occupied by rainfall intensities whose order of singularity exceeds the value γ . Schertzer and Lovejoy (1992) showed that Eq. (9) is valid in the space of statistical moments of the variable being analyzed (e.g., rainfall), which is equivalent to Eq. (8) in the probability space.

$$\langle e_{\lambda}^q \rangle = \lambda^{K(q)} \text{ for } \lambda > 1 \quad (9)$$

where q is the order of statistical moment; $\langle \rangle$ denotes expectation operator, $\langle e_{\lambda}^q \rangle$ is the average rainfall intensity at scale ratio; λ and $K(q)$ is the scaling exponent that is related to $c(\gamma)$ by the Legendre transform (Frisch, 1985):

$$\begin{aligned} c(\gamma) &= \max_q [q\gamma - K(q)] \\ K(q) &= \max_{\gamma} [q\gamma - c(\gamma)] \end{aligned} \quad (10)$$

Estimation of design precipitation using multifractal approach could be based on two options (Douglas and Barros, 2003): fractal maximum precipitation (FMP) and design probable maximum precipitation (DPMP). The estimate of FMP (Eq. (11)) is based on the observational record, whereas the estimate of DPMP (Eq. (12)) is based on a specified risk of exceedance. The design of a flood control structure in a river basin is based on PMF corresponding to DPMP estimates if the structure is likely to be exposed to a higher risk of failure. In other cases, FMP could be considered to arrive at a PMF estimate for the design of the structure.

$$\begin{aligned} \text{FMP} &= 10^B \tau^{1-\gamma_{max}} \\ \text{DPMP} &= 10^B \tau^{c(\gamma_e)} \end{aligned} \quad (11)$$

where B and γ_{max} are coefficients which can be computed from scaling relationship between maximum accumulations and their corresponding durations (Douglas and Barros, 2003); $c(\gamma_e)$ is codimension corresponding to the desired exceedance probability ρ_e (e.g., 10^{-6}), which is estimated using the following expression derived based on Eq. (10).

$$c(\gamma_e) = \ln(\rho_e^{-1}) / \ln(\lambda) \quad (12)$$

It is to be noted that FMP and DPMP estimates can be obtained using multifractal approach (MA) only for time scales that are longer than the shortest (e.g., daily) scale at which observations are available in the precipitation record. This is because when τ is equal to unity (i.e., 1 day), Eqs. (10), (11) indicate that estimates of FMP and DPMP would attain a constant value (10^B), irrespective of the values taken by γ_{max} and $c(\gamma_e)$.

3. Statistical PMP estimates: A case-study

3.1 Hershfield PMP estimates in Malaysia

PMP estimation in Malaysia are usually used for dam design works. Estimation of the PMP value uses the physical approach by maximizing major historic storm events. Storm transposition and envelopment techniques were incorporated to compensate for lack of adequate storm database (Desa et al., 2001). Due to lack of long observation records of meteorological data such as dew point temperature and wind speed, PMP estimates using the Hershfield statistical method was recommended as an option. Malaysia has enough rainfall data records for statistical Hershfield method but lack of dew point temperature and wind data. A study using the Hershfield PMP method was conducted by Desa et al. (2001), Desa and Rakhecha (2007), and Sidek et al. (2013). Desa et al. (2001) and Desa and Rakhecha (2007) uses the maximum value of the frequency factor, K_m to transposed the maximum rainfall for the PMP estimate. Table 2 shows the 24-h PMP value obtain for Johor state. The PMPs are in the range of 300–1200mm, with the ratio of the highest PMP against its highest recorded 24-h rainfall is approximately three. The rainfall stations used for the Johor state PMP estimation are stations within the state administrative boundary (see Fig. 3A) and has records between 20 and 55 years. The highest K_m value was applied to estimate the PMP. Sidek et al. (2013) however conducted an envelopment of the K_m values but also uses the maximum K_m value for the PMP estimates of all the stations within the Cameron Highland Batang Padang (CHBP) catchment boundary, Pahang. The 1-day, 3-day, and 5-day PMP maximum values are found to be 730.08, 966.17, and 969.0mm respectively with the largest ratio against the highest observed rainfall was 2.30.

Apart using stations within an administrative state or catchment boundary for the PMP statistical estimates, adopting rainfall stations within a homogenous region was attempted to compensate for limitation of rainfall observations records and, data quality (see Fig. 3A and B). Extreme-rainfall homogeneous regions refer to regions that contain sites with similar characteristics of extreme rainfall frequency distribution such as its means, skewness and kurtosis. This specifies that the areas within the homogeneous regions have similar frequency distribution characteristics and climatic exposure. L-moments regional frequency analysis was used to identify these homogeneous regions as it uses an application that is statistically efficient and straightforward to implement. A climatic homogeneous region developed by the Malaysia Department of Irrigation and Drainage was used to estimate the PMPs considering stations within a homogenous region. A total of 162 rainfall stations with 24-h rainfall data from the years 1969 to 2012 was used for the analysis. There are much more stations owned by DID; however, after data screening, only stations with more than 25 years of observation period and acceptable missing data were selected for the PMP analysis. Table 3 shows the results of the PMP values for stations within the homogenous region for region 2. A comparison of PMP estimated using administrative state boundary and the homogenous region is presented in Table 3 as well. It is found that the range of ratio of PMP values against its highest recorded rainfall is between 1 and 8. Further study on the limitation homogeneous boundary size area is needed to avoid large ratio values and overestimation of PMP estimates.

TABLE 2 Statistical 24-h PMP value estimated for the Johor State (Desa and Rakhecha, 2007).

Station number	$\bar{X}_n$ (mm)	σ_n (mm)	24-h PMP (mm)	24-h PMP/1-day highest
1	117	45	514	2.0
2	117	48	539	1.7
3	119	42	490	2.0
4	82	33	373	2.2
5	123	48	546	1.8
6	139	62	683	2.1
7	108	48	529	2.4
8	117	44	505	2.2
9	141	53	609	2.6
10	115	43	495	2.0
11	130	47	546	1.8
12	106	48	527	1.6
13	114	46	519	2.2
14	109	46	513	1.8
15	102	41	463	1.7
16	103	43	481	1.6
17	245	110	1209	2.2
18	117	48	539	1.8
19	86	34	385	2.8
20	104	42	473	1.9
21	111	42	481	1.9
22	113	53	577	1.7
23	156	76	820	2.9
24	100	33	393	1.8
25	96	33	389	1.6
26	104	40	457	1.9
27	128	53	594	2.2
28	134	64	694	2.4
29	117	58	624	2.6
30	96	34	397	1.8
31	101	37	428	1.7
32	127	63	678	2.1
33	167	58	680	2.5
34	139	66	716	2.1
35	200	85	947	2.4
36	115	54	588	1.9
37	210	67	805	2.1
38	106	59	620	2.2
39	222	70	844	1.9

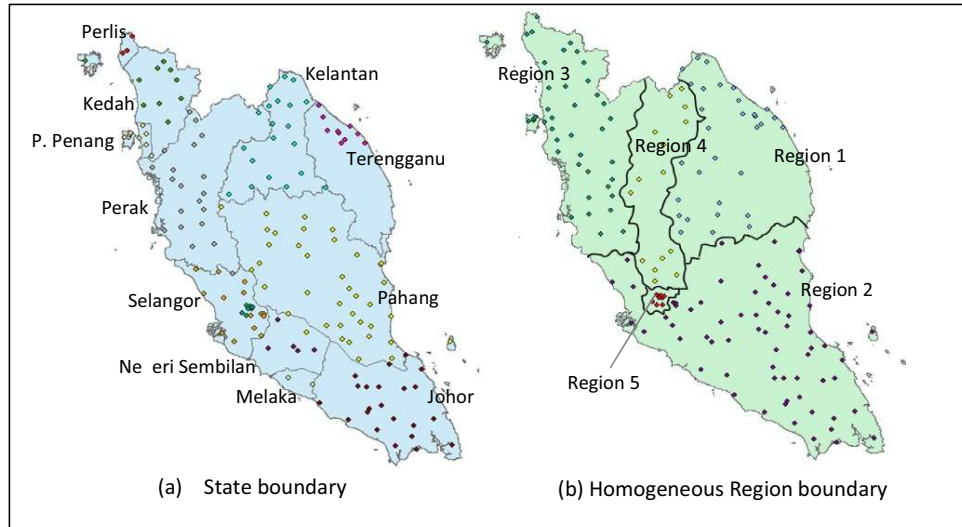


FIG. 3 Rainfall stations within a homogeneous region for the Peninsular Malaysia.

Another case study of statistical PMP estimates is presented as a station-wise PMP estimates in Bangladesh. For illustration purposes, example estimation of PMP is performed in this section. For this, the annual daily maximum rainfall of Sylhet meteorological station is collected from the Bangladesh Meteorological Department (BMD) (Table 4). Sylhet receives more than 4000mm of rainfall annually. It is the wettest region in Bangladesh. One other important aspect is that this region is highly susceptible to flash floods. Flash flood is often dominant by heavy rainfall in a short period of time. Thus, estimation of PMP will lead to a proper estimation of PMF.

Here

$$X_n = \frac{\sum \text{Daily maximum rainfall}}{\text{Number of year}} = \frac{8840}{44} = 200.91$$

$$S_n = 51.75$$

$$X_{max} = 362$$

$$X_{n-1} = 197.2$$

$$S_{n-1} = 45.93$$

Therefore,

$$K_m = \frac{X_{max} - X_{n-1}}{S_{n-1}} = \frac{362 - 197.2}{45.93} = 3.59$$

Hershfield noted in his study that the value of K_m is 3.5 for a 100 years of return period if the length of the record is 50 years (Hershfield, 1965a, b).

$$\text{And } X_{PMP} = X_n + S_n \times K_m = 200.91 + 51.75 \times 3.59 = 386.69$$

This example uses a K_m value obtained from the station itself instead from an envelope curve. The resulting return period estimated PMP is more than fifty years (Fig. 4). The calculation can be very useful especially for local projects (e.g., storm drainage, reservoir, etc.) in developing countries where PMF is calculated for less than hundred years. However, PMF for any return period can easily be calculated using Fig. 4.

The estimated PMP value of 386.69 almost coincides with that estimated by Bari (Bari, 2021). Bari (2021) estimated PMP using the Hershfield method for the whole Bangladesh. Estimated values are expected to give crude but quick information to investigate the adequacy of the hydraulic structures under extreme conditions and to estimate PMF for the selected region. The result indicates that although Sylhet is the wettest place in the country, one-day maximum rainfall is comparatively lower. This is maybe because the monsoon rain spell is characterized by a consistent but consecutive heavy rainy day. This consecutive heavy rainfall (one to five days) often causes catastrophic flash floods along with upstream hilly water.

TABLE 3 PMP estimates from state and homogeneous region boundary.

Station number	24-h highest rainfall (mm)	24-h PMP		Ratio PMP/highest rainfall record	
		State boundary	Region 2	State boundary	Region 2
1437116	285.3	405.2	731.6	1.42	2.56
1534002	198.5	336.2	626	1.69	3.15
1541139	406.4	695.4	993.2	1.71	2.44
1636001	250.5	445.2	875.7	1.78	3.50
1732004	215.5	351.3	707.2	1.63	3.28
1737001	298	471.7	877	1.58	2.94
1829002	198.5	421.2	983.2	2.12	4.95
1834001	327.5	483.8	886	1.48	2.71
1839196	397.5	667.3	976.8	1.68	2.46
1931001	149	295.2	996.6	1.98	6.69
1931003	167	313.1	616	1.87	3.69
2025001	187	337.8	619.7	1.81	3.31
2033001	204	369	610.1	1.81	2.99
2230001	185.4	342.4	700.9	1.85	3.78
2231001	202.9	385.6	751.9	1.90	3.71
2232001	331.1	589.3	954.2	1.78	2.88
2235163	433.5	581.9	739.8	1.34	1.71
2237164	355.9	620.9	816.1	1.74	2.29
2330009	310.7	477.4	918.7	1.54	2.96
2438001	311.5	660.1	1860.2	2.12	5.97
2534160	363.5	775.9	674	2.13	1.85
2636170	400.1	762.3	412.9	1.91	1.03
2224038	141.5	146	492.8	1.03	3.48
2321006	160	164.9	645.7	1.03	4.04
2719001	149.6	155.6	472.6	1.04	3.16
2722002	171.5	185.8	584.1	1.08	3.41
2723002	125	168.9	492.3	1.35	3.94
2725083	165.5	189	642.6	1.14	3.88
3020016	138.7	233.5	647.7	1.68	4.67
2815001	180.7	455.4	513.7	2.52	2.84
2818110	214.9	505.9	561.3	2.35	2.61
2913001	522.5	1229.1	1254.8	2.35	2.40
2917001	194	601.2	627.2	3.10	3.23
3117070	144.9	411.8	428.6	2.84	2.96
3118102	207.8	600.8	661.9	2.89	3.19
3119001	139.6	788.4	1060.5	5.65	7.60
3119002	109	594.6	958.4	5.46	8.79
3314001	193.5	590.7	642.2	3.05	3.32

Continued

TABLE 3 PMP estimates from state and homogeneous region boundary—cont'd

Station number	24-h highest rainfall (mm)	24-h PMP		Ratio PMP/highest rainfall record	
		State boundary	Region 2	State boundary	Region 2
3411017	190	552.5	601.6	2.91	3.17
3613004	121	1058.3	1032.7	8.75	8.53
2630001	423.9	1986.2	1097.1	4.69	2.59
2634193	472.9	2195.8	601.2	4.64	1.27
2828173	223	1135.4	843.4	5.09	3.78
2829001	235.3	1252.2	787.1	5.32	3.35
2831179	430.5	2102	1149.8	4.88	2.67
2834001	401.4	2184.5	926.7	5.44	2.31
2841001	339	1846.6	875	5.45	2.58
2924096	305.7	1219.6	929.3	3.99	3.04
3026156	362.6	1396.9	1084	3.85	2.99
3028001	181	782.4	568.1	4.32	3.14
3030178	228.8	1608.1	1212.9	7.03	5.30
3032167	298.5	1527.9	814.4	5.12	2.73
3121143	142	649.2	507.7	4.57	3.58
3129177	232.5	1383.6	939.4	5.95	4.04
3134165	410.8	1820.7	789.7	4.43	1.92
3228174	621.5	2260.5	1322.8	3.64	2.13
3231163	280	1276.6	719.8	4.56	2.57
3330109	305	1558.9	972.2	5.11	3.19
3424081	155.7	852.2	656.8	5.47	4.22
3429096	90.1	476.3	476.3	5.29	5.29
3533102	510	2273.9	760.9	4.46	1.49
3628001	317	1449.5	1021.8	4.57	3.22
3833001	423	2427	1867.8	5.74	4.42
3833002	433	1892.9	841.4	4.37	1.94
3924072	143.3	639.4	492.4	4.46	3.44
3930012	417.5	1876.4	1004.2	4.49	2.41

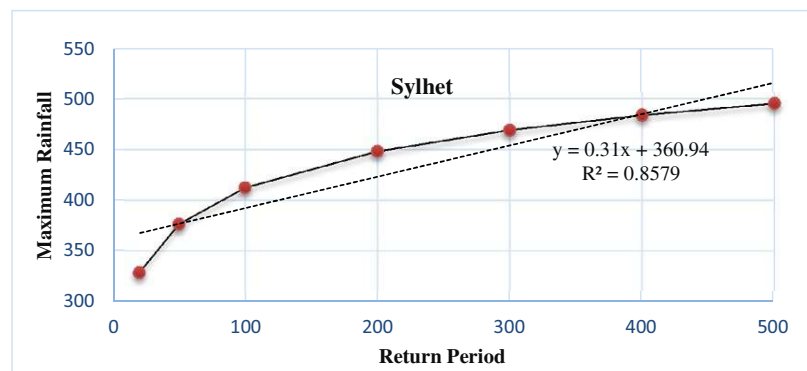
Station-wise PMP estimates in Bangladesh.

4. Conclusions

PMP is largely used for the estimation of the PMF and adequacy of different hydraulic structures. So, it is very important to estimate PMP as precisely as it can be. But, the greatest limitation of PMP estimation is the associated uncertainty. The definition of PMP itself is often dubious as such amount is probable, not possible. Furthermore, no allowance is made for long-term precipitation trend. The selection of method and a representative PMP estimation depends on several parameters. Physical based methods need several hydro climatological data which are often unavailable. Execution of these methods requires a certain level of expertise too. Also, they rely on some basic assumptions (Singh et al., 2018). In contrarily, the

TABLE 4 Daily maximum rainfall for Sylhet station of Bangladesh (44 years).

362	302	290	274	271	265	257	252	248	239	239
233	231	221	218	216	210	206	199	195	194	191
185	182	177	176	174	173	173	171	170	170	168
164	157	153	152	152	150	148	145	141	129	117

**FIG. 4** Maximum one day rainfall for different return period in Sylhet.

statistical approach is used where adequate precipitation data (at least 30–40 years) are available (Chavan and Srinivas, 2015; World Meteorological Organization, 2009a). These methods are based on frequency analysis which can be affected by the properties of data and method of computation. For example, Hershfield frequency factor (K_m) varies inversely with mean annual maximum rainfall at any station (Desa et al., 2001; World Meteorological Organization, 2009a). It is, therefore, important to calculate the frequency factor very carefully to minimize overestimation. However, a universally accepted calculation technique is still absent (Sarkar and Maity, 2020). The Multifractal method is constrained by the length of the record, the spatial resolution of the rain gauge network, and the lack of uncertainty estimates (Singh et al., 2018). The moisture Maximization Method is too sensitive to available data. Papalexiou and Koutsoyiannis (2006) argued that the method is flawed in its fundamental aspects. It is obvious each method has its advantages and disadvantages. So, it can be concluded that there is lack of a unique method for determining the upper bound of rainfall assuming it really exists (Koutsoyiannis, 1999). Therefore, attention should be given to selecting a particular method for a particular region.

Regardless of estimation methods, the PMP estimates are assumed to be free from any risk associated with their exceedance (Chavan and Srinivas, 2015). This leads to a larger return period of PMP events (100 years mostly). In many developing countries, this is unfavorable in terms of economy. Besides, despite of a considerably lower one-day PMP, consecutive rainy days in a similar manner should be taken into consideration in practice.

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